

LESSON 6-1 • SESSION 2 • PRACTICE • FORM A

Dividing a Fraction by a Whole Number

Learning target – I can divide a fraction by a whole number and explain the result. · 6.NS.A.1

REMEMBER

- Dividing by a whole number **shares** the fraction into equal parts.
- Multiply by the reciprocal** of the whole number:
 $2/3 \div 4 = 2/3 \times 1/4 = 2/12 = 1/6$.
- Each share is **less** than the fraction. Check: share $\times$ number of shares = the fraction.

WORKED EXAMPLE

A painter pours $2/3$ liter of paint evenly into 4 trays. How much paint is in each tray?

Step 1 Write the division: $2/3 \div 4$.

Step 2 Multiply by the reciprocal: $2/3 \times 1/4 = 2/12$.

Step 3 Simplify: $2/12 = 1/6$. Check: $1/6 \times 4 = 4/6 = 2/3$.

Answer $1/6$ liter in each tray

WORDS reciprocal (*recíproco*) · multiplicative inverse (*inverso multiplicativo*) · divisor (*divisor*)

PRACTICE

show your work · label units

1. Divide. Write each quotient in simplest form.

a. $1/3 \div 2 =$ _____

b. $3/5 \div 3 =$ _____

c. $4/5 \div 2 =$ _____

d. $7/10 \div 7 =$ _____

e. $8/9 \div 2 =$ _____

f. $5/6 \div 10 =$ _____

2. **Hiking Supplies.** A club splits $3/4$ pound of trail mix evenly among 4 hikers, as the tape shows.

$3/4$ lb

?	?	?	?
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- a. How much trail mix does each hiker get? _____ lb

- b. They pour $2/3$ gallon of water evenly into 10 bottles. Amount per bottle? _____ gal

3. **Clay Balls.** A sculptor has $3/5$ kilogram of clay. She divides it equally into 10 balls. What is the mass of each ball?

Hint Multiply $3/5$ by $1/10$.

Show your work.

Answer _____ kg

4. **Glue Bottles.** A jar holds $6/7$ liter of glue. It is poured evenly into 3 bottles. How much glue is in each bottle?

Hint Multiply by $1/3$, then simplify.

Show your work.

Answer _____ L

5. **String Pieces.** A $3/4$ -yard string is cut into 4 equal pieces. Priya says each piece is 3 yards long. Justify or correct her answer.

Hint Is one piece longer or shorter than the whole string?

Show your work.

Answer _____ yard

6. **Lemonade.** Sara has $\frac{7}{12}$ liter of lemonade. She pours out $\frac{3}{12}$ liter, then shares the rest equally among 4 friends.

Hint Subtract first, then share the rest.

- a. How much is left? _____ L
b. How much does each friend get? _____ L

Show your work.

7. **Baking Day.** A baker has $\frac{7}{8}$ kilogram of dough to split evenly among trays.

- a. How much dough goes on each tray if there are 7 trays? _____ kg
b. How much if there are 14 trays? _____ kg
c. How did doubling the number of trays change each share? _____

8. **Fabric Strip.** A strip of fabric is $\frac{8}{9}$ meter long. It is cut into 4 equal pieces. Find the length of one piece. Then check by multiplying.

Hint Check: your answer $\times$ 4 should equal $\frac{8}{9}$.

Show your work.

Answer _____ m

FIND THE ERROR

explain the mistake, then fix it

9. **Caleb's Work.** Caleb solved $\frac{3}{4} \div 10$. He kept the divisor right side up and wrote $\frac{3}{4} \times \frac{10}{1} = \frac{30}{4}$. Explain his mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

10. Find $\frac{1}{2} \div 4$.

A 2 **B** $\frac{1}{8}$ **C** 8 **D** $\frac{1}{2}$

11. The reciprocal of 8 is –

A $\frac{1}{8}$ **B** 8 **C** –8 **D** 0.8

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

6 X	3 Z	$\frac{2}{7}$ H	$\frac{2}{9}$ G	$\frac{1}{8}$ I	$\frac{1}{16}$ N	$\frac{1}{12}$ R	$\frac{3}{50}$ S	$\frac{18}{7}$ Q	$\frac{3}{16}$ A
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#3	#4	#5	#6b	#7a	#7b	#8
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____